

Problem 1: (15 Points)

Let $\mathcal{C}_{[0,1]} := \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ is continuous}\}$ and $d_\infty : \mathcal{C}_{[0,1]} \times \mathcal{C}_{[0,1]} \rightarrow \mathbb{R}_+$ be defined as

$$d_\infty(f, g) := \max_{t \in [0,1]} |f(t) - g(t)|.$$

- (a) (2 pts) For $f, g, h : [0, 1] \rightarrow \mathbb{R}$ with $f(x) = x$, $g(x) = x^2$ and $h(x) = 0.5x^2$, find $d_\infty(f, g)$ and $d_\infty(f, h)$.
- (b) (3 pts) Argue that d_∞ is a well-defined function. (Hint: You can use as fact that the absolute value is a continuous transformation.)
- (c) (10 pts) Show that $(\mathcal{C}_{[0,1]}, d_\infty)$ is a metric space.

Solution:

(a)

$$d_\infty(f, g) = \max_{x \in [0,1]} |x - x^2| = \max_{x \in [0,1]} (x - x^2).$$

Since $x - x^2$ is nonnegative on $[0, 1]$, its derivative is $1 - 2x$ which vanishes at $x = \frac{1}{2}$. Second derivative is a constant function $-2 < 0$ and thus it is indeed a maximum. Thus $d_\infty(f, g) = (\frac{1}{2} - \frac{1}{4}) = \frac{1}{4}$.

Similarly,

$$d_\infty(f, h) = \max_{x \in [0,1]} |x - \frac{1}{2}x^2| = \max_{x \in [0,1]} (x - \frac{1}{2}x^2).$$

The derivative is $1 - x$, which vanishes at $x = 1$. Second derivative is a constant function $-1 < 0$ and thus it is indeed a maximum. Hence $d_\infty(f, h) = 1 - \frac{1}{2} = \frac{1}{2}$.

- (b) For $f, g \in \mathcal{C}_{[0,1]}$, the function $f - g$ is continuous on $[0, 1]$. The absolute value map is continuous, so $|f - g|$ is continuous on $[0, 1]$. Because $[0, 1]$ is compact, the Weierstrass extreme value theorem guarantees that a continuous function attains a (finite) maximum on it. Therefore $\max_{t \in [0,1]} |f(t) - g(t)|$ exists and lies in $\mathbb{R}_+$, so d_∞ is well-defined.
- (c) Let $f, g, h \in \mathcal{C}_{[0,1]}$.

- Nonnegativity and identity of indiscernibles: $d_\infty(f, g) \geq 0$ by definition. If $d_\infty(f, g) = 0$, then $|f(t) - g(t)| = 0$ for all t , so $f = g$. Conversely, if $f = g$, then $d_\infty(f, g) = 0$.
- Symmetry: $d_\infty(f, g) = \max_t |f(t) - g(t)| = \max_t |g(t) - f(t)| = d_\infty(g, f)$.
- Triangle inequality: For each $t \in [0, 1]$,

$$|f(t) - h(t)| \leq |f(t) - g(t)| + |g(t) - h(t)|.$$

For any two real-valued functions A, B on $[0, 1]$, we have

$$\max_{t \in [0,1]} (A(t) + B(t)) \leq \max_{t \in [0,1]} A(t) + \max_{t \in [0,1]} B(t).$$

Indeed, for each t ,

$$A(t) + B(t) \leq \max_{u \in [0,1]} A(u) + \max_{v \in [0,1]} B(v),$$

and taking the maximum over t on the left preserves the inequality.
Applying this with $A(t) = |f(t) - g(t)|$ and $B(t) = |g(t) - h(t)|$ yields

$$\max_{t \in [0,1]} (|f(t) - g(t)| + |g(t) - h(t)|) \leq \max_{t \in [0,1]} |f(t) - g(t)| + \max_{t \in [0,1]} |g(t) - h(t)|.$$

Thus d_∞ satisfies all metric axioms, and $(\mathcal{C}_{[0,1]}, d_\infty)$ is a metric space.

Problem 2: (25 points)

Let $\rho : \mathbb{R} \times \mathbb{R} \rightarrow \{0, 1\}$ be the discrete metric, i.e.

$$\rho(x, y) := \begin{cases} 1, & \text{if } x \neq y, \\ 0, & \text{if } x = y \end{cases}$$

- (a) (5 pts) Show that any singleton in $\mathbb{R}$ (i.e. any set made up of a single point $\{x\} \subset \mathbb{R}$) is open in $(\mathbb{R}, \rho)$.
- (b) (5 pts) Show that any subset of $\mathbb{R}$ is open in $(\mathbb{R}, \rho)$.
- (c) (5 pts) Show that any subset of $\mathbb{R}$ is closed in $(\mathbb{R}, \rho)$.
- (d) (5 pts) Show that any subset of $\mathbb{R}$ is bounded in $(\mathbb{R}, \rho)$.
- (e) (5 pts) Show that $[0, 1]$ is not compact in $(\mathbb{R}, \rho)$.
- (f) (**Bonus:** 5 pts) Show that the sequence $(1/n)$ does not converge to 0 in $(\mathbb{R}, \rho)$.

Solution:

- (a) Fix $x \in \mathbb{R}$. Consider the open ball of radius $1/2$ around x :

$$B_{1/2}(x) = \{y \in \mathbb{R} : \rho(x, y) < 1/2\}.$$

Since $\rho(x, y)$ takes only the values 0 or 1, the inequality $\rho(x, y) < 1/2$ forces $y = x$. Thus $B_{1/2}(x) = \{x\}$, so every singleton is open.

- (b) Let $U \subset \mathbb{R}$. For any $x \in U$, we know $B_{1/2}(x) = \{x\} \subset U$. Hence every point of U is interior, so U is open. Equivalently,

$$U = \bigcup_{x \in U} \{x\},$$

a union of open sets.

- (c) A set is closed if its complement is open. Since by part (b) every subset of $\mathbb{R}$ is open, the complement of any subset is also open. Thus every subset of $\mathbb{R}$ is both open and closed (clopen).
- (d) A subset $S \subset \mathbb{R}$ is bounded if there exists $a \in \mathbb{R}$ and $M < \infty$ such that $\rho(a, x) \leq M$ for all $x \in S$. Since $\rho(a, x) \leq 1$ for all $a, x \in \mathbb{R}$, we can take $M = 1$, which shows every subset of $\mathbb{R}$ is bounded in the discrete metric.

- (e) Consider the open cover

$$\mathcal{U} = \{\{x\} : x \in [0, 1]\}.$$

Each $\{x\}$ is open (by part (a)) and the union of all such sets is $[0, 1]$. But any finite subcollection of $\mathcal{U}$ covers only finitely many points, never the whole interval $[0, 1]$. Hence $[0, 1]$ has an open cover with no finite subcover and is therefore not compact. In fact, in the discrete metric, compact sets are exactly the finite sets.

- (f) (**Bonus**) Suppose the sequence $x_n = 1/n$ converges to 0. Then pick $\varepsilon = \frac{1}{2}$, there must exist N such that for all $n \geq N$, $\rho(1/n, 0) < 1/2$. But for every n , $1/n \neq 0$, so $\rho(1/n, 0) = 1$. This is not less than $1/2$, hence the convergence condition fails. Therefore $(1/n)$ does not converge to 0. More generally, in the discrete metric a sequence converges to x if and only if it is eventually constant equal to x .

Problem 3: (15 points)

Define $\mathbf{1}_n \in \mathbb{R}^n$ as a column vector of size n full of ones, i.e. $\mathbf{1}_n := (1, 1, \dots, 1)$. Now define $S_n := \mathbf{1}_n \mathbf{1}_n^T$, for any $n \in \mathbb{N}$.

- (a) (3 pts) Find the eigenvalues of S_2 and S_3 .
- (b) (7 pts) Find $\sigma(S_n)$ for $n > 3$. (Hint: Find a pattern from your answer to part (a), and use row operations that preserve determinant.)
- (c) (5 pts) In general, is S_n diagonalizable in $\mathbb{R}$? If so, propose a spectral decomposition; that is, find an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$. (Note: You do *not* have to compute P^{-1} .)

Next, a bonus question about vectors unrelated to (a)–(c):

- (d) (**Bonus:** 5 pts) If v_1, v_2, v_3 are L.I. non-zero vectors in some vector space $(V, +, \cdot)$, is $\{v_1, 2v_1 + v_3, 4v_2 + 2v_3\}$ a basis of $\text{span}(v_1, v_2, v_3)$?

Solution:

- (a) For $S_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$,

$$\lambda I - S_2 = \begin{pmatrix} \lambda - 1 & -1 \\ -1 & \lambda - 1 \end{pmatrix} \xrightarrow{C_2 \leftarrow C_2 - C_1} \begin{pmatrix} \lambda - 1 & -\lambda \\ -1 & \lambda \end{pmatrix}.$$

Factoring λ from column 2 gives

$$\det(\lambda I - S_2) = \lambda \det \begin{pmatrix} \lambda - 1 & -1 \\ -1 & 1 \end{pmatrix} = \lambda(\lambda - 2).$$

Thus the eigenvalues are 0 and 2.

For $S_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$,

$$\lambda I - S_3 = \begin{pmatrix} \lambda - 1 & -1 & -1 \\ -1 & \lambda - 1 & -1 \\ -1 & -1 & \lambda - 1 \end{pmatrix} \xrightarrow{C_2 \leftarrow C_2 - C_1, C_3 \leftarrow C_3 - C_1} \begin{pmatrix} \lambda - 1 & -\lambda & -\lambda \\ -1 & \lambda & 0 \\ -1 & 0 & \lambda \end{pmatrix}.$$

Factoring λ from columns 2 and 3, and then adding rows 2 and 3 to row 1,

$$\det(\lambda I - S_3) = \lambda^2 \det \begin{pmatrix} \lambda - 1 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \lambda^2(\lambda - 3).$$

Thus the eigenvalues are 0, 0, 3.

- (b) For general n , start with $A = \lambda I - S_n$, which has diagonal entries $\lambda - 1$ and off-diagonal entries -1 . For $j = 2, \dots, n$ perform the column operation $C_j \leftarrow C_j - C_1$. Each such column then has only two nonzero entries: $(1, j) = -\lambda$ and $(j, j) = \lambda$. Factoring λ from each of columns $2, \dots, n$ gives

$$\det(\lambda I - S_n) = \lambda^{n-1} \det \begin{pmatrix} \lambda - 1 & -1 & \cdots & -1 \\ -1 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ -1 & 0 & \cdots & 1 \end{pmatrix}.$$

Now add rows $2, \dots, n$ to row 1, obtaining the first row $(\lambda - n, 0, \dots, 0)$ and a lower block equal to I_{n-1} . Thus

$$\det(\lambda I - S_n) = \lambda^{n-1}(\lambda - n).$$

Therefore the eigenvalues are n (once) and 0 (with multiplicity $n - 1$).

- (c) Since S_n is real symmetric ($S_n = S_n^\top$), it is orthogonally diagonalisable by the spectral theorem. Hence there exists an orthogonal matrix Q with

$$Q^\top S_n Q = \text{diag}(n, 0, \dots, 0).$$

Since orthogonal diagonalisability implies diagonalisability, we now move to proposing an invertible (not necessarily orthogonal) matrix P .

To see how to construct such a matrix explicitly, note that

$$\text{Im}(S_n) = \text{span}\{\mathbf{1}_n\}, \quad \ker(S_n) = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0\},$$

and these two subspaces intersect at only the zero vector, so that

$$\mathbb{R}^n = \text{span}\{\mathbf{1}_n\} \oplus \ker(S_n).$$

Choose the basis consisting of $\mathbf{1}_n$ together with the vectors $e_k - e_n$ for $k = 1, \dots, n - 1$. Then in this basis the matrix of S_n is $\text{diag}(n, 0, \dots, 0)$. Thus one may take

$$P = [\mathbf{1}_n \mid e_1 - e_n \mid e_2 - e_n \mid \cdots \mid e_{n-1} - e_n],$$

and then $P^{-1}S_nP = \text{diag}(n, 0, \dots, 0)$.

- (d) Relative to the independent list (v_1, v_2, v_3) , the coordinate matrix with columns $w_1 = v_1$, $w_2 = 2v_1 + v_3$, $w_3 = 4v_2 + 2v_3$ is

$$M = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 2 \end{pmatrix}.$$

Its determinant is

$$\det M = 1 \cdot \det \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} - 2 \cdot \det \begin{pmatrix} 0 & 4 \\ 0 & 2 \end{pmatrix} = -4 \neq 0.$$

Hence w_1, w_2, w_3 are linearly independent. Since each w_i belongs to $\text{span}(v_1, v_2, v_3)$, they form a basis of this space.

Alternatively, to test whether $\{v_1, 2v_1 + v_3, 4v_2 + 2v_3\}$ is a basis of $\text{span}(v_1, v_2, v_3)$, assume

$$av_1 + b(2v_1 + v_3) + c(4v_2 + 2v_3) = 0 \implies (a + 2b)v_1 + (4c)v_2 + (b + 2c)v_3 = 0.$$

Since v_1, v_2, v_3 are independent, the coefficients satisfy

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Row-reduce to solve:

$$\begin{aligned} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 2 \end{pmatrix} &\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{pmatrix} \\ &\xrightarrow{R_3 \leftarrow \frac{1}{4}R_3} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 4R_3, R_2 \leftarrow R_2 - 2R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Thus the unique solution is $a = b = c = 0$, so the three vectors are linearly independent. Since each lies in $\text{span}(v_1, v_2, v_3)$ (dimension 3), they form a basis of this space.

Problem 4: (10 points)

Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(u_1, u_2, u_3) = \begin{bmatrix} \ln(u_1 + u_2^2 + u_3^3) \\ u_1^2 \end{bmatrix}, \quad g(v_1, v_2) = \begin{bmatrix} e^{v_1} \\ \sqrt{v_2} \end{bmatrix}, \quad h(w_1, w_2) = w_1^2 - w_2^2.$$

- (a) (2 pts) Compute the Jacobian matrices of f, g and h .
 (b) (3 pts) Compute the partial derivatives:

$$\frac{\partial(h \circ g \circ f)}{\partial u_i}$$

for $i = 1, 2, 3$.

- (c) (5 pts) Approximate the value of $h \circ g \circ f(1.1, 1.9, -0.8)$ using a first order Taylor approximation. (Hint: Use $(1, 2, 1)$ as reference point).
 (d) (**Bonus:** 5 pts) Approximate the value of $h \circ g \circ f(1.1, 1.9, -0.8)$ using a second order Taylor approximation.

Solution:

- (a) Let $S = u_1 + u_2^2 + u_3^3$. For the Jacobians, the derivative of f is

$$Df(u) = \begin{pmatrix} \frac{1}{S} & \frac{2u_2}{S} & \frac{3u_3^2}{S} \\ 2u_1 & 0 & 0 \end{pmatrix},$$

for g we obtain

$$Dg(v) = \begin{pmatrix} e^{v_1} & 0 \\ 0 & \frac{1}{2\sqrt{v_2}} \end{pmatrix},$$

and for h we have

$$Dh(w) = (2w_1 \quad -2w_2).$$

- (b) By the chain rule,

$$D(h \circ g \circ f)(u) = Dh(g(f(u))) Dg(f(u)) Df(u).$$

Since $f(u) = (\ln S, u_1^2)$, one has $g(f(u)) = (S, u_1)$. Therefore

$$Dh(g(f(u))) = (2S, -2u_1).$$

Multiplying with $Dg(f(u)) = \text{diag}(S, 1/(2u_1))$ yields

$$Dh(g(f(u))) Dg(f(u)) = (2S^2, -1).$$

Finally,

$$\nabla(h \circ g \circ f)(u) = (2S^2, -1) Df(u) = (2(u_2^2 + u_3^3), 4Su_2, 6Su_3^2).$$

Hence

$$\frac{\partial}{\partial u_1}(h \circ g \circ f) = 2(u_2^2 + u_3^3), \quad \frac{\partial}{\partial u_2}(h \circ g \circ f) = 4Su_2, \quad \frac{\partial}{\partial u_3}(h \circ g \circ f) = 6Su_3^2.$$

- (c) At the reference point $u_0 = (1, 2, 1)$ we have $S_0 = 1 + 4 + 1 = 6$. Evaluating the composite function gives

$$f(u_0) = (\ln 6, 1), \quad g(f(u_0)) = (6, 1), \quad h(g(f(u_0))) = 36 - 1 = 35.$$

The gradient at this point is

$$\nabla(h \circ g \circ f)(u_0) = (10, 48, 36).$$

For the nearby point $(1.1, 1.9, -0.8)$, the displacement from u_0 is $\Delta u = (0.1, -0.1, -1.8)$. A first-order Taylor approximation gives

$$h \circ g \circ f(1.1, 1.9, -0.8) \approx 35 + (10)(0.1) + (48)(-0.1) + (36)(-1.8) = 35 - 68.6 = -33.6.$$

- (d) For the second-order approximation it is convenient to note that $g(f(u)) = (S, u_1)$ for $u_1 > 0$, hence

$$(h \circ g \circ f)(u) = S^2 - u_1^2 = (u_1 + u_2^2 + u_3^3)^2 - u_1^2.$$

This function has first derivatives $H_{u_1} = 2(u_2^2 + u_3^3)$, $H_{u_2} = 4Su_2$, $H_{u_3} = 6Su_3^2$, and second derivatives

$$H_{11} = 0, \quad H_{12} = 4u_2, \quad H_{13} = 6u_3^2, \quad H_{22} = 4S + 8u_2^2, \quad H_{23} = 12u_2u_3^2, \quad H_{33} = 18u_3^4 + 12Su_3.$$

At $u_0 = (1, 2, 1)$ these evaluate to

$$H_{11} = 0, \quad H_{12} = 8, \quad H_{13} = 6, \quad H_{22} = 56, \quad H_{23} = 24, \quad H_{33} = 90.$$

The Hessian is

$$\nabla^2 H(u) = \begin{pmatrix} 0 & 4u_2 & 6u_3^2 \\ 4u_2 & 4S + 8u_2^2 & 12u_2u_3^2 \\ 6u_3^2 & 12u_2u_3^2 & 18u_3^4 + 12Su_3 \end{pmatrix}.$$

At $u_0 = (1, 2, 1)$ with $S_0 = 6$, this becomes

$$\nabla^2 H(u_0) = \begin{pmatrix} 0 & 8 & 6 \\ 8 & 56 & 24 \\ 6 & 24 & 90 \end{pmatrix}.$$

The quadratic correction is

$$\frac{1}{2} \Delta u^\top \nabla^2 H(u_0) \Delta u = \frac{1}{2}(-0.16 - 2.16 + 0.56 + 8.64 + 291.6) = 149.24.$$

Therefore the second-order Taylor expansion yields

$$h \circ g \circ f(1.1, 1.9, -0.8) \approx 35 - 68.6 + 149.24 = 115.64.$$

Problem 5: (15 points)

Find the maximizers of the following problem

$$\max_{(x,y,z) \in \mathbb{R}^3} f(x, y, z) = -x^2 + 2x - y^2 + 4y - z^2 + 6z \quad \text{subject to} \quad x \leq 2, \quad y + 2z \geq 12$$

Solution:

We want to maximize

$$f(x, y, z) = -x^2 + 2x - y^2 + 4y - z^2 + 6z$$

subject to $x \leq 2$ and $y + 2z \geq 12$. Rewrite the constraints in KKT form as $c_1(x) = 2 - x \geq 0$ and $c_2(y, z) = y + 2z - 12 \geq 0$. The Lagrangian is

$$\mathcal{L}(x, y, z, \mu_1, \mu_2) = f(x, y, z) + \mu_1(2 - x) + \mu_2(y + 2z - 12), \quad \mu_1, \mu_2 \geq 0.$$

Stationarity (first-order conditions) is

$$\begin{aligned} -2x + 2 - \mu_1 &= 0, \\ -2y + 4 + \mu_2 &= 0, \\ -2z + 6 + 2\mu_2 &= 0. \end{aligned} \tag{S}$$

Complementary slackness is

$$\begin{aligned}\mu_1(2-x) &= 0, \\ \mu_2(y+2z-12) &= 0.\end{aligned}\tag{CS}$$

Primal feasibility requires $2-x \geq 0$, $y+2z-12 \geq 0$, and dual feasibility requires $\mu_1, \mu_2 \geq 0$.

From (S) we have $\mu_1 = 2-2x$, $\mu_2 = 2y-4$, and $z = 2y-1$.

If $x = 2$, then $\mu_1 = -2 < 0$, which violates dual feasibility. Therefore the first constraint cannot be active. By (CS) this forces $\mu_1 = 0$. Substituting into (S) gives

$$x^* = 1.$$

For the second constraint, suppose it were slack. Then $\mu_2 = 0$, and from (S) we would have

$$y = 2, \quad z = 2y - 1 = 3.$$

This point violates feasibility since

$$y + 2z - 12 = 2 + 6 - 12 = -4 < 0.$$

Hence the second constraint must be active:

$$y + 2z - 12 = 0.$$

Using $z = 2y - 1$, this becomes

$$y + 2(2y - 1) - 12 = 0 \implies 5y - 14 = 0.$$

Thus

$$y^* = \frac{14}{5}, \quad z^* = 2y^* - 1 = \frac{23}{5}.$$

The corresponding multiplier is

$$\mu_2^* = 2y^* - 4 = \frac{8}{5} > 0.$$

Therefore the KKT point is

$$(x^*, y^*, z^*) = \left(1, \frac{14}{5}, \frac{23}{5}\right),$$

and

$$f(x^*, y^*, z^*) = \frac{54}{5}.$$

Problem 6: (20 points)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : \mathbb{R} \rightrightarrows Y$ has a compact graph iff the graph of F is a compact set in metric space $(X \times Y, d_{X \times Y})$ where

$$d_{X \times Y}((x, y), (x', y')) := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}.$$

Use as given the fact that x_n converges to x in (X, d_X) and y_n converges to y in (Y, d_Y) if and only if (x_n, y_n) converges to (x, y) in $(X \times Y, d_{X \times Y})$.

- (5 pts) Show that if F has a compact graph then F is compact-valued.
- (5 pts) Show that the converse may not hold, i.e. F may be compact-valued but still not have a compact graph.
- (5 pts) Show that if F has a compact graph then F is locally bounded at any $x_0 \in X$.
- (5 pts) If F has a compact graph, is it true that F would be uhc? Explain.

Solution:

- (a) Assume that the graph $\text{Gr}(F)$ is compact in $X \times Y$. Fix any point $x \in X$. The slice $\{x\} \times Y$ is closed in the product metric space $X \times Y$, because in metric spaces singletons are always closed, the whole space is both closed and open and products of closed sets are closed. Therefore the intersection

$$K_x = \text{Gr}(F) \cap (\{x\} \times Y)$$

is compact since closed subset of a compact set is compact. By construction we have

$$K_x = \{(x, y) : y \in F(x)\}.$$

Now consider the projection map $\pi_Y : X \times Y \rightarrow Y$, defined by $\pi_Y(x, y) = y$. This map is continuous, and $\pi_Y(K_x) = F(x)$. Since the image of a compact set under a continuous map is compact, it follows that $F(x)$ is compact for every $x \in X$. This shows that if the graph is compact then the correspondence is compact-valued.

- (b) The converse is not true in general. As a counterexample, let $X = Y = \mathbb{R}$ with the usual metric, and define $F(x) = \{x\}$ for each $x \in \mathbb{R}$. Each $F(x)$ is a singleton, and singletons are compact in $\mathbb{R}$. Hence F is compact-valued. But the graph of F is

$$\text{Gr}(F) = \{(x, x) : x \in \mathbb{R}\},$$

which is the diagonal line in $\mathbb{R}^2$. This set is closed, but it is not bounded, and therefore by the Heine–Borel theorem it is not compact in $\mathbb{R}^2$. Thus it is possible to have a compact-valued correspondence without a compact graph.

- (c) Fix $x_0 \in X$ and $r > 0$. The closed ball $\overline{B}_X(x_0, r)$ is closed in X , hence $\overline{B}_X(x_0, r) \times Y$ is closed in $X \times Y$. Intersecting with the compact graph yields

$$G_r := \text{Gr}(F) \cap (\overline{B}_X(x_0, r) \times Y),$$

which is compact (closed subset of a compact set is compact). The projection π_Y is continuous, thus

$$B_r := \pi_Y(G_r) = \{y \in Y : \exists x \in \overline{B}_X(x_0, r) \text{ with } (x, y) \in \text{Gr}(F)\} = \bigcup_{x \in \overline{B}_X(x_0, r)} F(x)$$

is compact in Y as the image of a compact set under a continuous map. Compactness implies boundedness. (Note that for this simplification we don't need Heine–Borel so we are not assuming Y to be Euclidean.) Consequently, for every x with $d_X(x, x_0) \leq r$ we have $F(x) \subset B_r$, which is exactly local boundedness at x_0 .

- (d) Take any sequence $x_n \rightarrow x_0$ in X and for each n pick some $y_n \in F(x_n)$. Then the points (x_n, y_n) all belong to $\text{Gr}(F)$. Because the graph is compact and thus sequentially compact, the sequence (x_n, y_n) has a convergent subsequence, say $(x_{n_k}, y_{n_k}) \rightarrow (x, y)$. Convergence in the product metric means that $x_{n_k} \rightarrow x$ in X and $y_{n_k} \rightarrow y$ in Y . Since we also know $x_{n_k} \rightarrow x_0$, it must be that $x = x_0$. Now, because the graph is closed, the limit point (x_0, y) lies in $\text{Gr}(F)$. This means that $y \in F(x_0)$. In other words, every sequence $x_n \rightarrow x_0$ and $y_n \in F(x_n)$ has subsequences with limit points in $F(x_0)$. This is exactly the sequential definition of upper hemicontinuity. Therefore a compact graph implies that F is upper hemicontinuous.

A quicker argument is to use the following proposition:

Proposition: If F has a closed graph and is locally bounded at x_0 , then F is upper hemicontinuous at x_0 .

Since a compact graph is closed (this is true for every metric space) and (by part (c)) F is locally bounded at every $x_0 \in X$, it follows from the proposition that F is upper hemicontinuous.